

Week 11 – List Algorithms

■ Learning Objectives

- Learn and apply sorting algorithms (Bubble Sort, Built-in Sort).
- Understand linear and binary search methods.
- Explore sublists, slicing, and merging lists.
- Compare algorithm efficiency and problem-solving patterns.

■ Key Concepts

1. Sorting arranges elements; searching finds specific values.
2. Linear search checks each item one by one.
3. Binary search halves the search space each time (requires sorting).
4. Bubble sort demonstrates the logic of comparison-based sorting.
5. Python's built-in sort (Timsort) is optimized and preferred.

Example 1 – Bubble Sort

```
nums = [5, 3, 8, 4, 2]
n = len(nums)
for i in range(n-1):
    for j in range(n-i-1):
        if nums[j] > nums[j+1]:
            nums[j], nums[j+1] = nums[j+1], nums[j]
print(nums)
```

Example 2 – Linear Search

```
nums = [10, 20, 30, 40]
target = 30
for i in range(len(nums)):
    if nums[i] == target:
        print("Found at index", i)
        break
else:
    print("Not found")
```

Example 3 – Binary Search

```
nums = [1, 3, 5, 7, 9, 11]
target = 9
low, high = 0, len(nums)-1
while low <= high:
    mid = (low + high) // 2
    if nums[mid] == target:
        print("Found at index", mid)
        break
    elif nums[mid] < target:
        low = mid + 1
    else:
        high = mid - 1
```

Example 4 – Merge Two Sorted Lists

```
a = [1, 3, 5]
b = [2, 4, 6]
merged = []
i = j = 0
while i < len(a) and j < len(b):
    if a[i] < b[j]:
        merged.append(a[i])
        i += 1
    else:
        merged.append(b[j])
        j += 1
merged.extend(a[i:])
merged.extend(b[j:])
print(merged)
```

■ Practice Problems

Practice 1 – Second Largest Number

```
nums = [4, 9, 1, 7, 3]
nums.sort()
print("Second largest:", nums[-2])
```

Practice 2 – Remove Duplicates

```
nums = [1, 2, 2, 3, 4, 4, 5]
unique = []
for n in nums:
    if n not in unique:
        unique.append(n)
print(unique)
```

Practice 3 – Average of Sublist

```
nums = [10, 20, 30, 40, 50]
start = int(input("Start index: "))
end = int(input("End index: "))
sub = nums[start:end+1]
print("Average:", sum(sub)/len(sub))
```

■ Efficiency Comparison

Python's `sorted()` uses Timsort ($O(n \log n)$), which is much faster than bubble sort ($O(n^2)$). For CCC-level problems, always prefer built-in sorting unless demonstrating algorithm logic.

■ Homework Challenge

Write a program that reads a list of integers and prints the 3 smallest and 3 largest unique values.

```
nums = list(map(int, input("Enter numbers: ").split()))
unique = sorted(set(nums))
print("Smallest 3:", unique[:3])
print("Largest 3:", unique[-3:])
```